

## IZMIR CIGLI, Turkey

WMO: 172180

Lat: 38.513N

Lon: 27.014E

Elev: 16

StdP: 14.69

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 29.8       | 31.9 | 17.4                          | 13.2 | 35.3 | 19.8 | 14.9 | 37.6 | 25.4                  | 54.0 | 23.0 | 53.5 | 4.8                   | 0    | 0.469 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 23.6                   | 98.4            | 70.6 | 95.4 | 70.3 | 93.4 | 69.7 | 74.6                | 91.0 | 73.2 | 89.7 | 72.0 | 88.4 | 13.3                 | 350  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 69.6                            | 109.0 | 83.0 | 67.9 | 102.5 | 81.8 | 66.2 | 96.6 | 80.7 | 38.1          | 91.4 | 36.8 | 89.4 | 35.6 | 88.8 | 89.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |     |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |     |
|-------------------|------|------|----------------------------|-------|-----|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|-----|
| 1%                |      | 2.5% | 5%                         | Mean  |     | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |     |
| Min               | Max  | Min  | Max                        | Min   | Max | Min                | Max                                                | Min       | Max  | Min        | Max  | Min        | Max  | Min        | Max |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g) | (h)                | (i)                                                | (j)       | (k)  | (l)        | (m)  | (n)        | (o)  | (p)        |     |
| 22.2              | 19.9 | 18.0 | 24.7                       | 103.8 | 2.9 | 3.7                | 22.6                                               | 106.4     | 20.9 | 108.5      | 19.3 | 110.6      | 17.1 | 113.2      |     |
|                   |      |      | 22.7                       | 77.7  | 3.3 | 3.5                | 20.4                                               | 80.2      | 18.5 | 82.3       | 16.6 | 84.2       | 14.3 | 86.7       |     |
|                   |      |      | DB                         |       |     |                    |                                                    |           |      |            |      |            |      |            |     |
|                   |      |      | WB                         |       |     |                    |                                                    |           |      |            |      |            |      |            |     |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 63.9   | 46.9  | 48.8  | 54.0  | 60.3  | 69.3  | 77.3  | 81.9  | 81.6  | 73.6  | 65.6  | 56.9  | 49.7  |
|                                                                        | DBStd       | 13.59  | 7.33  | 7.13  | 6.33  | 5.48  | 4.74  | 4.97  | 3.31  | 3.63  | 4.35  | 5.63  | 6.56  | 7.40  |
|                                                                        | HDD50       | 395    | 148   | 99    | 32    | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 14    | 100   |
|                                                                        | HDD65       | 2323   | 560   | 454   | 342   | 160   | 16    | 0     | 0     | 0     | 1     | 61    | 253   | 476   |
|                                                                        | CDD50       | 5477   | 53    | 66    | 157   | 312   | 598   | 818   | 989   | 981   | 709   | 483   | 221   | 90    |
|                                                                        | CDD65       | 1932   | 0     | 1     | 2     | 19    | 149   | 369   | 524   | 516   | 260   | 79    | 11    | 2     |
|                                                                        | CDH74       | 21923  | 0     | 1     | 12    | 184   | 1378  | 4326  | 6679  | 6313  | 2452  | 548   | 30    | 0     |
|                                                                        | CDH80       | 10437  | 0     | 0     | 1     | 29    | 379   | 2091  | 3625  | 3331  | 895   | 84    | 2     | 0     |
| (14) Wind                                                              | WSAvg       | 7.8    | 7.7   | 8.3   | 8.4   | 7.4   | 7.5   | 8.3   | 9.5   | 8.9   | 7.5   | 6.7   | 6.6   | 7.5   |
| (15) Precipitation                                                     | PrecAvg     | 25.4   | 4.1   | 3.7   | 2.8   | 2.1   | 1.4   | 0.6   | 0.2   | 0.2   | 0.7   | 1.9   | 3.3   | 4.4   |
|                                                                        | PrecMax     | 38.3   | 8.7   | 9.5   | 6.4   | 4.7   | 5.2   | 1.9   | 0.8   | 1.5   | 3.1   | 6.9   | 9.8   | 9.1   |
|                                                                        | PrecMin     | 15.6   | 0.2   | 0.4   | 0.7   | 0.4   | 0.1   | 0.0   | 0.0   | 0.0   | 0.1   | 0.0   | 0.1   | 0.1   |
|                                                                        | PrecStd     | 5.5    | 2.4   | 2.5   | 1.5   | 1.1   | 1.1   | 0.5   | 0.2   | 0.3   | 0.9   | 1.5   | 2.0   | 2.5   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 66.0  | 69.4  | 75.3  | 83.6  | 90.4  | 100.6 | 102.2 | 101.0 | 94.6  | 85.8  | 77.4  | 70.1  |
|                                                                        |             | MCWB   | 54.5  | 55.9  | 58.9  | 62.6  | 65.5  | 70.0  | 71.6  | 71.4  | 68.5  | 66.1  | 63.8  | 57.9  |
|                                                                        | 2%          | DB     | 62.8  | 64.8  | 70.1  | 78.9  | 87.4  | 96.7  | 98.4  | 97.2  | 90.0  | 82.4  | 73.3  | 66.3  |
|                                                                        |             | MCWB   | 54.2  | 54.3  | 56.3  | 60.8  | 65.2  | 69.4  | 70.9  | 71.2  | 67.6  | 65.1  | 61.5  | 57.1  |
|                                                                        | 5%          | DB     | 60.7  | 62.5  | 67.8  | 75.3  | 84.3  | 93.3  | 95.4  | 95.1  | 87.8  | 79.2  | 70.2  | 64.0  |
|                                                                        |             | MCWB   | 53.0  | 53.8  | 55.4  | 59.6  | 64.4  | 68.5  | 70.3  | 71.1  | 67.0  | 63.5  | 59.8  | 56.3  |
|                                                                        | 10%         | DB     | 57.6  | 59.8  | 64.7  | 71.8  | 81.2  | 90.4  | 93.4  | 93.1  | 85.6  | 77.0  | 67.9  | 60.9  |
|                                                                        |             | MCWB   | 51.2  | 52.2  | 54.1  | 57.8  | 63.5  | 67.6  | 69.7  | 70.5  | 66.7  | 62.6  | 58.6  | 54.6  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 57.5  | 57.4  | 60.1  | 65.5  | 69.6  | 73.7  | 76.8  | 76.9  | 72.9  | 69.4  | 65.7  | 60.0  |
|                                                                        |             | MCDB   | 63.5  | 65.4  | 71.0  | 78.5  | 84.5  | 91.1  | 93.5  | 92.7  | 86.2  | 80.2  | 74.0  | 66.5  |
|                                                                        | 2%          | WB     | 55.1  | 55.6  | 58.0  | 63.0  | 67.7  | 71.8  | 74.5  | 75.0  | 70.9  | 67.2  | 63.3  | 58.3  |
|                                                                        |             | MCDB   | 61.2  | 63.3  | 67.2  | 74.6  | 82.2  | 89.1  | 91.5  | 91.0  | 83.9  | 77.7  | 70.8  | 64.5  |
|                                                                        | 5%          | WB     | 53.2  | 54.1  | 56.5  | 60.9  | 66.2  | 70.4  | 73.0  | 73.6  | 69.5  | 65.7  | 61.1  | 56.6  |
|                                                                        |             | MCDB   | 59.2  | 61.4  | 65.1  | 72.4  | 80.4  | 87.6  | 90.1  | 89.7  | 82.6  | 76.0  | 68.5  | 62.9  |
|                                                                        | 10%         | WB     | 51.7  | 52.6  | 55.1  | 59.2  | 64.8  | 69.1  | 71.8  | 72.3  | 68.1  | 64.3  | 59.4  | 54.8  |
|                                                                        |             | MCDB   | 57.5  | 59.2  | 63.6  | 69.8  | 78.4  | 86.3  | 89.0  | 88.5  | 81.5  | 74.5  | 66.7  | 60.4  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 14.1  | 15.3  | 17.1  | 20.2  | 22.0  | 23.2  | 23.6  | 23.1  | 22.6  | 20.4  | 17.5  | 14.1  |
|                                                                        |             | MCDBR  | 14.1  | 17.4  | 19.8  | 25.6  | 26.3  | 27.4  | 26.5  | 25.3  | 25.8  | 22.9  | 18.8  | 12.9  |
|                                                                        |             | MCWBR  | 9.2   | 10.6  | 11.1  | 12.9  | 11.1  | 9.6   | 9.8   | 9.1   | 10.5  | 10.7  | 10.7  | 8.4   |
|                                                                        | 5% WB       | MCDBR  | 12.8  | 15.7  | 17.5  | 22.5  | 23.4  | 24.2  | 23.8  | 22.9  | 22.8  | 20.4  | 16.6  | 12.0  |
|                                                                        |             | MCWBR  | 9.0   | 10.0  | 10.6  | 12.4  | 10.8  | 9.5   | 9.8   | 9.3   | 10.4  | 10.7  | 10.2  | 8.6   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.361 | 0.394 | 0.431 | 0.470 | 0.445 | 0.410 | 0.407 | 0.413 | 0.405 | 0.401 | 0.378 | 0.366 |
|                                                                        | taud        |        | 2.375 | 2.252 | 2.142 | 2.064 | 2.200 | 2.338 | 2.346 | 2.320 | 2.338 | 2.328 | 2.353 | 2.372 |
|                                                                        | Ebn at Noon |        | 254   | 259   | 261   | 258   | 267   | 276   | 275   | 272   | 266   | 254   | 245   | 242   |
|                                                                        | Edh at Noon |        | 29    | 36    | 45    | 51    | 46    | 40    | 39    | 39    | 36    | 33    | 29    | 27    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 651   | 888   | 1341  | 1761  | 2161  | 2444  | 2524  | 2237  | 1762  | 1235  | 822   | 598   |
|                                                                        | RadStd      |        | 77    | 89    | 96    | 161   | 106   | 125   | 48    | 48    | 92    | 69    | 70    | 85    |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (14 neighbors) | +0.68 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -167  | N/A   | N/A   |

Nomenclature: See separate page